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The Editors
Journal of Cheminformatics
BioMed Central

Dear Editors,

We are pleased to submit our manuscript, **“Pareto-guided Monte Carlo tree search explores multi-objective trade-offs missed by scalar antimalarial generation”**, for consideration as a Research Article in the *Journal of Cheminformatics*.

The work addresses a recurring limitation of de novo molecular design: scalar reward aggregation that conceals trade-offs between potency, synthetic accessibility and drug-likeness. We introduce a Pareto-guided Monte Carlo tree search (MCTS) agent that maintains a non-dominated front across multi-parameter optimisation (MPO), synthetic Bayesian accessibility (SYBA), resistance-resilience (RRS) and polypharmacology-network (PNS) scores, guided by a scaffold-aware fragment policy.

A four-method, 20-seed benchmark under a fixed search budget yields a transparent ranking: random search achieves the highest mean scalar reward (0.7335), followed by MCTS at its screened optimal configuration (0.7276; mean gap 0.0059, paired $t_{19} = 2.41$, $p = 0.026$), deterministic greedy search (0.7211) and a genetic algorithm (0.7027). MCTS is therefore competitive on single-objective performance and, under the multi-objective protocol, returns a Pareto front of non-dominated candidates (four solutions, hypervolume 1.2366) spanning the potency–accessibility trade-off, including a high-potency / low-accessibility extreme that scalar optimisation misses. Component ablations identify the chemistry-informed policy and the Pareto front itself as the dominant performance drivers.

We believe this transparent, methods-focused contribution aligns closely with the scope of the *Journal of Cheminformatics*: rigorous benchmarking, open deposition of code and molecule sets, and honest reporting of both strengths and limitations of a multi-objective search strategy. All data and code are released under the MIT licence and are immediately available to reviewers in the public repository (https://github.com/NanaEngo/Malaria_codesV2) and on Zenodo (reserved DOI <https://doi.org/10.5281/zenodo.19608875>).

This work is original, is not under consideration elsewhere, and all authors have approved the submission.

We thank you for your consideration.

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Suggested reviewers:

- Dr. Jan H. Jensen (graph-based genetic algorithms / MCTS for chemical space)

- Prof. Gisbert Schneider (generative molecular design)
- Dr. Ola Engkvist (de novo design, multi-objective optimisation)